hw-Lab2

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7. Interpretive questions:

1. **What do the most positive keywords (overrepresented in 2025) tell you about how data center coverage has shifted?**

The biggest difference in 2025 compared to 2021 is the high frequency of 'Trump' and 'AI'. In 2025, there is also words like 'tariffs', 'nuclear', 'bannon', 'power', and 'energy'. These all show the shift towards politics and energy politics instead of technical infrastructure. Today, data center coverage is being covered through the AI boom lens as well as seen as an expansion to the political environment. Additionally, there is a common story line displaying data centers demand for energy.

2. **Are any of these keywords surprising? What might explain their appearance?**

'altman', 'marcos', 'bannon' and 'doutthat' are surprising to me.

- Sam Altman is the CEO of Open AI.
- Marcos might refers to the San Marcos Data Center project. The $1.5 billion, 200 megawatt project was highly protested within the community and was rejected in the San Marcos City Council in February of 2026. Or it might refer to the Phillippines' Marcos administration as the Phillippines become a regional data center hub.
- Steve Bannon was employed as the White House chief strategist under Trump.
- Douthat is a New York Times opinion columnist.

I am surprised these names occurred enough to flag them. This illustrates that coverage is becoming more personality-driven and political.

3. **Keyness flags statistical unusualness, not substantive importance. A high-G^2 word could be driven by a single prolific source or a short news spike. How would you check whether a keyword is genuinely distributed across the corpus rather than concentrated in one place?**

You can check by investigating overall frequency, frequency by source, and monthly frequency. This checks distribution of words on various levels: source, overtime, and overall, making sure the words are not concentrated in one place.

4. **How has the language used to describe data centers in news coverage changed over time, and what does that suggest about the evolving journalistic frame around the issue?**

In 2021, the language is more localized, technical, and business oriented. In 2025, the language shifts towards more national contentment and energy politics. The language has changed from being about "where and who is building the infrastructure" to "how is the infrastructure impacting political and electrical power".

But, we have to be careful when using Keyness because it only shows the relative change. There can be similarities between time frames, but they are not shown because they are stable.

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8. Exercises

Stopword sensitivity: Remove *all* domain-specific stopwords and re-run the frequency plot. Then add 10 more domain stopwords beyond what was provided. How do your results change? What does this tell you about the analytical choices embedded in preprocessing?

Frequency plot when removing all domain-specific stopwords from class

```
# Remove all domain-specidic stopwords
domain_sw <- tibble(word = c(
# News wire / attribution boilerplate
"said", "says", "told", "according", "reported", "noted", "added",
"stated", "announced", "mr", "ms", "inc", "corp", "llc",
# Generic quantitative terms ##(adjust based on your research question)
"year", "years", "percent", "million", "billion", "quarter",
"company", "companies", "firm", "business",
# Consider whether these are substantive for our RQ:
"data", "center", "centers"
  ))

# Remove domain-specidic stopwords
tokens_clean <- tokens_clean |>
  anti_join(domain_sw, by = "word") |>
  filter(
    !str_detect(word, "^[0-9]+$"),    #remove pure digit tokens... No pure numbers
    str_length(word) > 2            #remove very short tokens
  )

  # Re-Run the frequency plot
top_words <- tokens_clean |>
  count(word, sort = TRUE) |>
  slice_head(n = 30)

frequency_plot1 <- top_words |>
  ggplot(aes(x = reorder(word, n), y = n)) +
  geom_col(fill = "#1a9641") +
  coord_flip() +
  labs(
    title = "30 Most Frequent Terms\nData Center Corpus",
    subtitle = "Class Stopwords",
    x = NULL,
    y = "Frequency"
  ) +
  theme_minimal(base_size = 13)
```

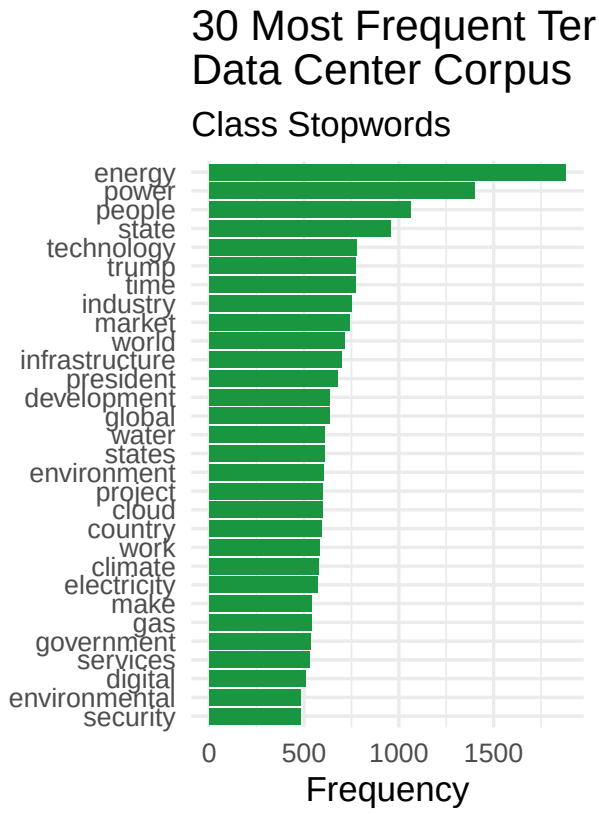
Frequency plot - Adding 10 more domain stopwords beyond what was provided

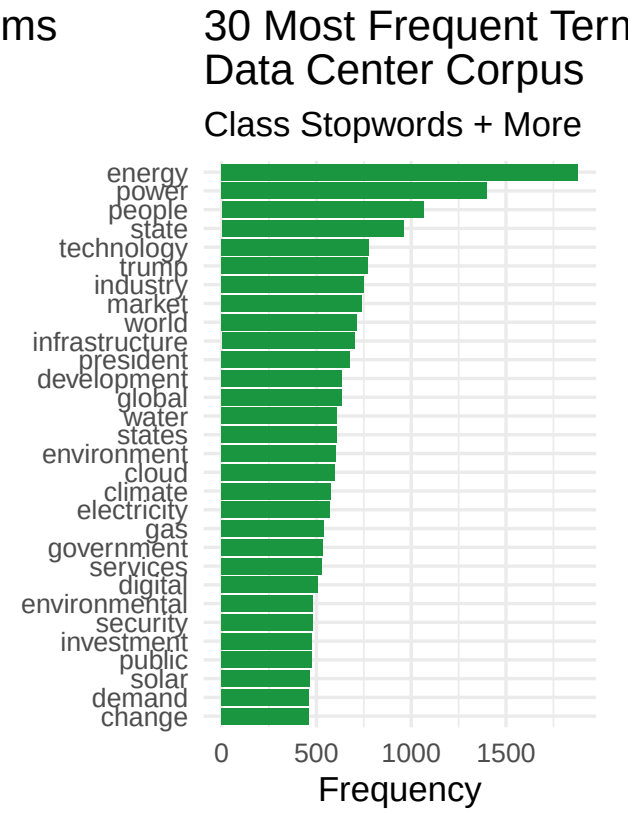
```
# Remove all domain-specidic stopwords + 10 more
domain_sw <- tibble(word = c(
# News wire / attribution boilerplate
"said", "says", "told", "according", "reported", "noted", "added",
"stated", "announced", "mr", "ms", "inc", "corp", "llc",
# Generic quantitative terms ##(adjust based on your research question)
"year", "years", "percent", "million", "billion", "quarter",
"company", "companies", "firm", "business",
# Consider whether these are substantive for our RQ:
```

```r
"data", "center", "centers",
# Adding 10 more words
"make", "time", "published", "devine", "memoir", "loving", "mind", "inspired", "drawn", "study", "family

  # Remove domain-specidic stopwords
tokens_clean2 <- tokens_clean |>
  anti_join(domain_sw, by = "word") |>
  filter(
    !str_detect(word, "^[0-9]+$"),
    #remove pure digit tokens... No pure numbers
    str_length(word) > 2            #remove very short tokens
  )

# Re-Run the frequency plot
top_words2 <- tokens_clean2 |>
  count(word, sort = TRUE) |>
  slice_head(n = 30)

frequency_plot2 <- top_words2 |>
  ggplot(aes(x = reorder(word, n), y = n)) +
  geom_col(fill = "#1a9641") +
  coord_flip() +
  labs(
    title = "30 Most Frequent Terms\nData Center Corpus",
    subtitle = "Class Stopwords + More",
    x = NULL,
    y = "Frequency"
  ) +
  theme_minimal(base_size = 13)
```

Comparing the Frequency plots

```r
# Compare the 2 plots
frequency_plot1 | frequency_plot2
```





All decisions during the preprocessing stage must be met with reason and deliberate thought. Removing vs not removing specific words impact in the results. For instance, the first frequency plot includes words like 'project' and 'make'. Whereas in the second frequency, I removed these words. Preprocessing has a direct impact on the results and must be treated with thought.

Alternative keyness comparison: Instead of comparing time periods, compare the two sources with the most articles. What does keyness reveal about how each publication frames the data center beat?

CE Noticias Financieras English and New York Times has the most articles

```
# See who the top news sources are
top_articles <- df %>%
  group_by(source) %>%
  summarise(count = n()) %>%
  arrange(desc(count)) %>%
  slice(1:2)

top_articles
```

```
## # A tibble: 2 x 2
##   source                          count
##   <chr>                           <int>
## 1 The New York Times                 71
## 2 CE Noticias Financieras English    67
```

Building a corpus and DFM

```r
# Build corpus for the sources
corp2 <- corpus(df, docid_field = "id", text_field = "text")

# Cleaning Tokenize inside quanteda: remove punctuation, numbers, symbols; lowercase
toks <- tokens(corp2,
       remove_punct   = TRUE,
       remove_numbers = TRUE,
       remove_symbols = TRUE) %>%
  tokens_tolower() |>
  tokens_remove(stopwords("en", source = "smart"))

# Build DFM
dfmat <- dfm(toks)
# Drop any documents with no source
dfmat <- dfmat[!is.na(docvars(dfmat, "source")), ]
```

```r
# Look at DFM features
cat("DFM dimensions:", nrow(dfmat), "documents ×", ncol(dfmat), "features\n")
```

```
## DFM dimensions: 528 documents × 25603 features
```

Define target and reference sub-corpora

Top two articles: New York Times and CE Noticias Financieras English

```r
# Define Groups
cat("Top source:", top_articles$source[1], "with", sum(docvars(dfmat, "source") == "The New York Times")
```

```
## Top source: The New York Times with 71 articles
```

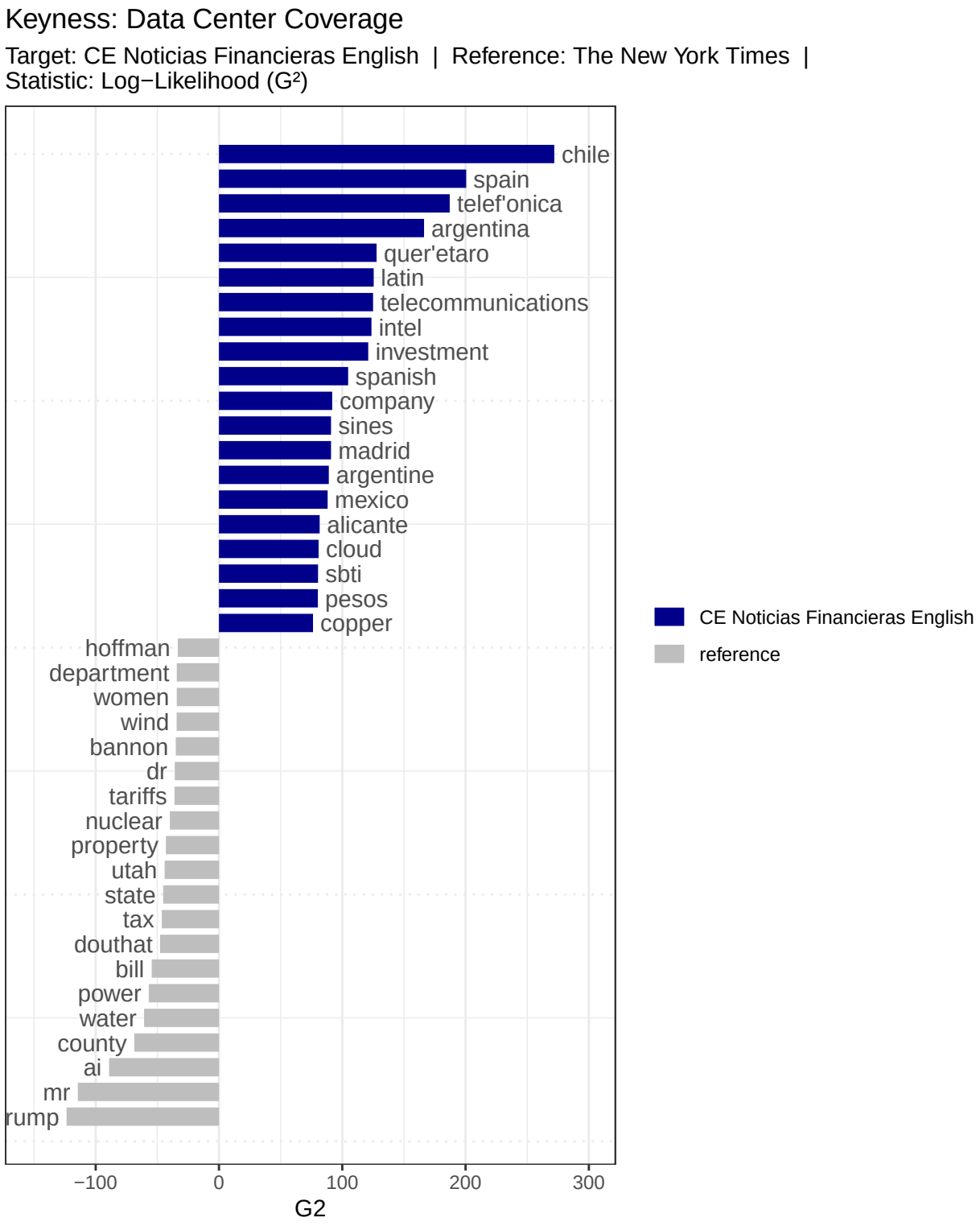
```r
cat("Second-top source:", top_articles$source[2], sum(docvars(dfmat, "source") == "CE Noticias Financie
```

```
## Second-top source: CE Noticias Financieras English 67 articles
```

```r
# Collapse DFM to two rows - one per year value
dfmat_grouped <- dfm_group(dfmat, groups = source)
```

Compute and plot keyness

```r
result <- textstat_keyness(dfmat_grouped,
                           target  = "CE Noticias Financieras English",
                           measure = "lr")

textplot_keyness(result, n = 20) +
  labs(
    title = "Keyness: Data Center Coverage",
    subtitle = "Target: CE Noticias Financieras English  |  Reference: The New York Times  |\nStatistic
  )
```



The CE Noticias Financieras coverage is dominated by Latin American geography and actors, including Chile, Spain, Argentina, Querétaro, Telefónica, pesos. The CE Noticias Financieras' data center must focused around regional infrastructure stories. The New York Times coverage is dominated by U.S policy and politics, including Trump, tariffs, Bannon, wind, and nuclear. Unlike the CE Noticias Financieras, NYT's data center focuses on domestic political controversy. This displays how the keyness can reflect the publication's ideology

and/or focus, showing their interpretive lens.

Bigrams: Use `unnest_tokens(token = "ngrams", n = 2)` to extract bigrams. What are the 20 most frequent bigrams? Do they reveal entities or conceptual phrases that single-token analysis misses?

Most of the common bigrams are function words with little or no independent meaning on their own, such as 'of the' and 'in the'. The most conceptual phrase is 'data center(s)'. Therefore, the 20 most common bigrams do not reveal anything important in this study.

```r
# Get the top 20 most frequen
df %>%
  unnest_tokens(bigram, text, token = "ngrams", n = 2) %>%
  count(bigram, sort = TRUE) %>%
  slice_head(n = 20)
```

```
## # A tibble: 20 x 2
##    bigram           n
##    <chr>        <int>
##  1 of the        3532
##  2 in the        3034
##  3 to the        1704
##  4 data centers  1301
##  5 data center   1223
##  6 and the       1167
##  7 for the       1140
##  8 on the        1095
##  9 to be          964
## 10 at the         816
## 11 with the       816
## 12 in a           753
## 13 that the       714
## 14 is a           667
## 15 as a           615
## 16 of a           595
## 17 u s            591
## 18 the company    589
## 19 according to   571
## 20 from the       568
```